

Labwork 6: Map

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1 Introduction

This report presents three image-processing operations implemented: grayscale binarization, brightness control, and blending two images on GPU.

2 Method

For an array A and a function f , the Map pattern is defined as

$$\text{map}(A, f) = [f(x_i), \forall x_i \in A] \quad (1)$$

A 16×16 thread block is used for all three labworks. Test image: a 3840×2160 RGB image.

3 Grayscale Binarization

Each pixel's RGB value is first averaged into a grayscale intensity $\Phi(x, y)$, then thresholded against $\tau = 128$:

$$b(x, y) = \begin{cases} 0 & \Phi(x, y) < \tau \\ 255 & \Phi(x, y) \geq \tau \end{cases} \quad (2)$$

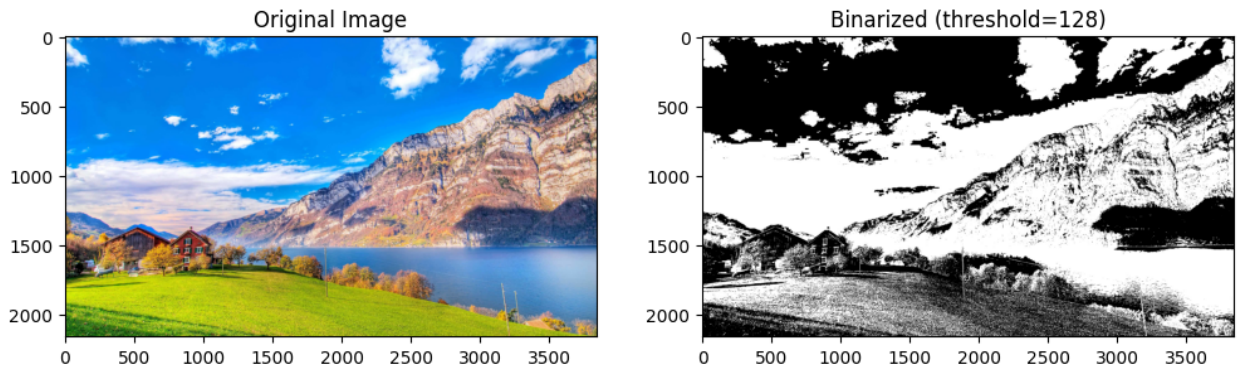


Figure 1: Original image vs. binarized output

4 Brightness Control

Each channel of each pixel is shifted by a constant and clamped back into the valid $[0, 255]$ range:

$$I'(x, y) = \min(\max(I(x, y) + \beta, 0), 255) \quad (3)$$

with $\beta = 50$.

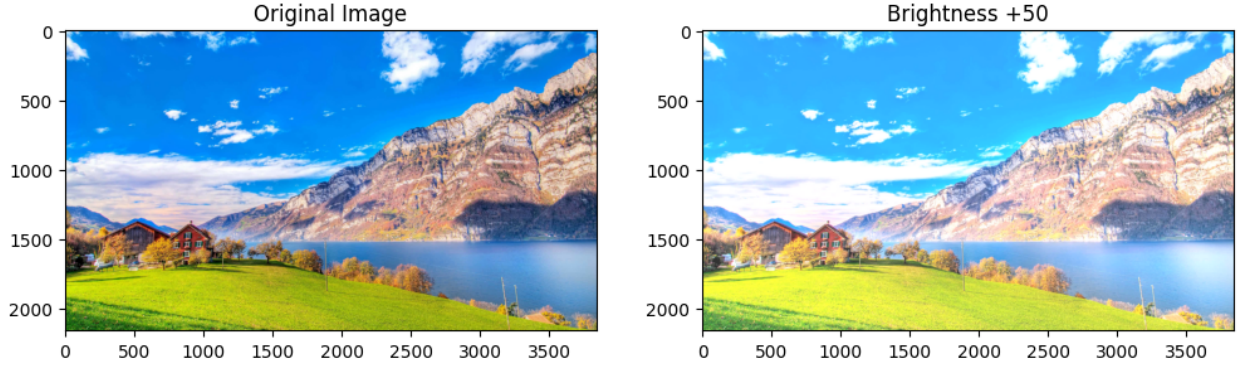


Figure 2: Original image (left) vs. brightness adjusted output

5 Blending Two Images

Two same-size images Φ_1 and Φ_2 are combined per-channel using a weight c :

$$Q(x, y) = c \times \Phi_1(x, y) + (1 - c) \times \Phi_2(x, y) \quad (4)$$

The second image was generated by horizontally flipping the original ($c = 0.5$).

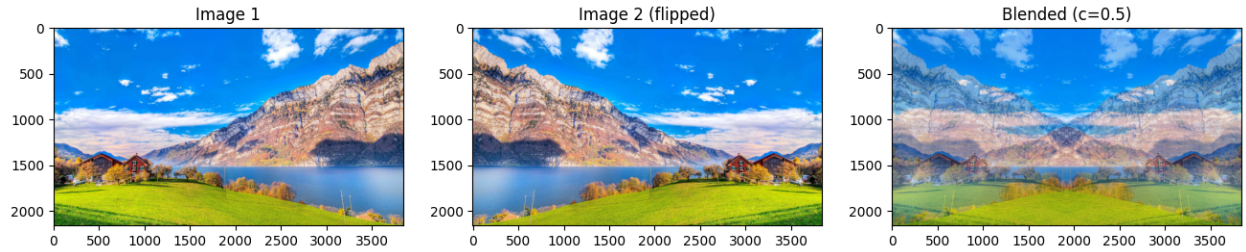


Figure 3: Image 1 (left), Image 2 / flipped

6 Results

Labwork	GPU time (s)
6a: Binarization	2.1619
6b: Brightness control	0.1451
6c: Blending	0.2114

Table 1: Execution time.